

ATPL EASA STUDY GUIDE | 070 - OPERATIONAL PROCEDURES

Chapter 5: Extended Diversion Time Operations (ETOPS / EDTO)

Syllabus Source: CAE Oxford (pp. 157-196) | Pass Target: $\geq 90\%$ AviationExam

1. THE ETOPS REGULATORY CONCEPT

ETOPS (Extended-range Twin-engine Operational Performance Standards) / EDTO permits twin-engine aircraft to fly long-range routes across oceans and deserts:

- The 60-Minute Threshold: Without ETOPS approval, a twin-jet must remain within 60 MINUTES FLYING TIME at One-Engine-Inoperative (OEI) cruise speed in still air from an adequate aerodrome.
- ETOPS Approval Ratings: ETOPS 120 min, 180 min, or 240+ min. The approved diversion time defines a circle around en-route alternates: $\text{Radius (NM)} = \text{Approved ETOPS Time (hours)} \times \text{Approved OEI Cruise TAS (knots)}$!

2. ADEQUATE VS SUITABLE AERODROME

- Adequate Aerodrome: Physically capable: Runway length, strength, emergency services (RFFS), and lighting meet aircraft requirements.
- Suitable Aerodrome (ETOPS En-Route Alternate): An adequate aerodrome whose FORECAST WEATHER AT EXPECTED TIME OF ARRIVAL (ETA) is AT OR ABOVE ETOPS DISPATCH MINIMA (which includes safety buffers above normal landing minima)!

STEP-BY-STEP PROBLEM SOLVING (FOR DUMMIES / PASO A PASO)

[+] STEP-BY-STEP ALGORITHM / PASO A PASO - Step-by-Step Algorithm (Dummies Guide): ETOPS Threshold Distance & Altitude

Problem: A twin-jet has an approved single-engine cruise speed of 420 kt TAS in standard atmosphere. Find the threshold distance beyond which ETOPS regulations apply. Step 1: ETOPS threshold is 60 minutes (1.0 hour) single-engine cruise speed. Step 2: Threshold Distance = $420 \text{ kt} \times 1.0 \text{ h} = 420 \text{ NM}$! Step 3: If any point on the route is $> 420 \text{ NM}$ from an adequate airport, ETOPS approval is mandatory. Step 4: If operator holds ETOPS 180 min approval: Maximum allowable diversion distance is $420 \text{ kt} \times 3.0 \text{ h} = 1,260 \text{ NM}$.

3. STEP-BY-STEP WORKED EXAMPLE (FOR DUMMIES / PASO A PASO)

[*] OFFICIAL DEFINITION - PRACTICAL CALCULATION: Calculating ETOPS 120-Minute Maximum Diversion Range

SCENARIO (Transatlantic Route Planning Drill): An Airbus A330 operator has ETOPS 120-minute approval: - Approved One-Engine Inoperative (OEI) cruise speed in still air = 420 knots TAS QUESTION: What is the maximum allowable diversion distance (radius of ETOPS circle) in Nautical Miles from an en-route alternate aerodrome? SOLUTION PASO A PASO (FOR DUMMIES): Step 1: Identify Approved Diversion Time and OEI Speed: - ETOPS Time = 120 minutes = 2.0 hours. - Approved OEI Speed = 420 knots TAS. Step 2: Recall the ETOPS Maximum Diversion Distance Formula: - Rule: ETOPS distance is calculated strictly in STILL AIR (zero wind credit/penalty)! - Formula: Maximum Range (NM) = Diversion Time (hours) x OEI TAS Step 3: Calculate the Maximum Range: - Range = 2.0 hours x 420 knots = 840 Nautical Miles! Step 4: Operational Meaning for Dummies: - The aircraft must never be further than 840 NM from a suitable ETOPS alternate at any point along the ocean route! FINAL ANSWER: Maximum allowable diversion distance is exactly 840 Nautical Miles.

[!] EXAM TRAP (AVIATIONEXAM) - The Still Air ETOPS Distance Trap

- The maximum ETOPS diversion distance is calculated in STILL AIR (zero wind)! - Even if there is a 100 knot jet stream tailwind, you CANNOT stretch the 840 NM circle! - The circle radius is fixed by: Approved Hours x OEI TAS in still air.

4. AVIATIONEXAM PRACTICE QUESTIONS (WITH EXPLANATIONS)**Q1: What is the baseline threshold diversion time for twin-turbine aeroplanes beyond which ETOPS / EDTO apply?**

- [A]: 30 minutes
- [B]: 60 MINUTES at one-engine-inoperative cruise speed in still air
- [C]: 120 minutes
- [D]: 180 minutes

[?] OFFICIAL RATIONALE & ANALYSIS - Detailed Answer Explanation

CORRECT: [B]. Any twin-engine commercial flight route that exceeds 60 minutes diversion time to an adequate aerodrome is classified as an extended diversion time operation (ETOPS).

Q2: In ETOPS dispatch regulations, what distinguishes a 'SUITABLE AERODROME' from an 'ADEQUATE AERODROME'?

- [A]: Only runway length
- [B]: A suitable aerodrome must have forecast weather at ETA meeting or exceeding higher ETOPS DISPATCH MINIMA
- [C]: Suitable aerodromes must be military
- [D]: There is no difference

[?] OFFICIAL RATIONALE & ANALYSIS - Detailed Answer Explanation

CORRECT: [B]. An adequate aerodrome meets physical criteria; a suitable aerodrome also has confirmed weather forecasts meeting strict ETOPS dispatch buffers during the arrival window.

Q3: An operator has 180-minute ETOPS approval. If their approved OEI cruise speed is 400 kt TAS in still air, what is the maximum allowable diversion distance (radius of ETOPS circle) in Nautical Miles?

- [A]: 800 NM
- [B]: 1,200 NAUTICAL MILES (3.0 hours x 400 kt = 1,200 NM)

- [C]: 400 NM
- [D]: 1,800 NM

[?] OFFICIAL RATIONALE & ANALYSIS - Detailed Answer Explanation

CORRECT: [B]. 180 minutes = 3 hours. 3 hours x 400 knots = 1,200 NM maximum distance from an en-route alternate.